Customer Shopping Behavior Analysis

Uncovering insights from 3,900 transactions to guide strategic business decisions

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Dataset Overview

3,900

Transactions

Purchase records analyzed

18

Features

Data points per transaction

37

Missing Values

Review ratings imputed

Dataset includes customer demographics, purchase details, and shopping behavior metrics. Key features encompass age, gender, location, subscription status, item purchased, category, purchase amount, season, size, color, discount applied, promo code usage, previous purchases, purchase frequency, review rating, and shipping type.

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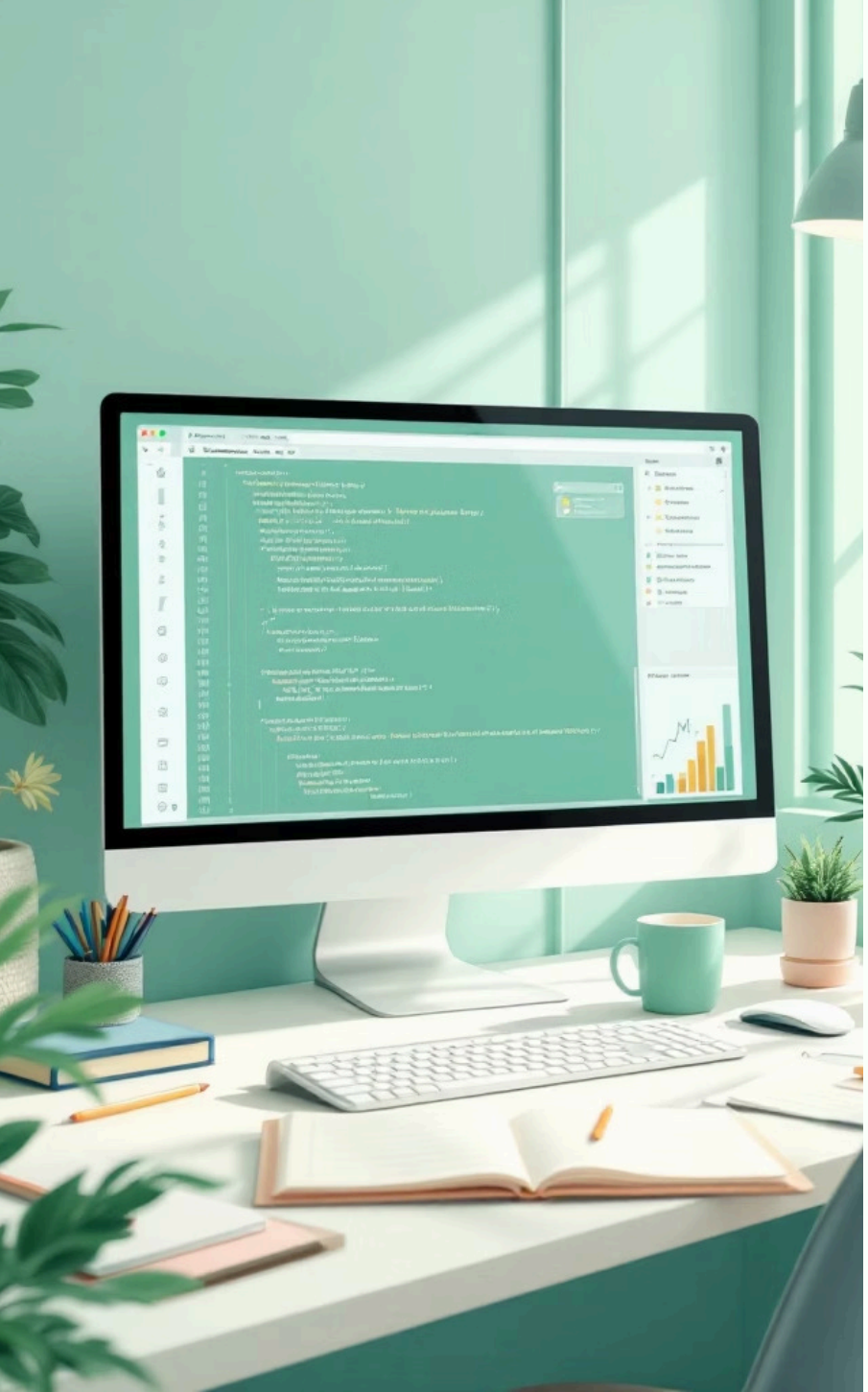

Python Data Preparation

Data Loading & Exploration

Imported dataset using `pandas`. Used `df.info()` to check structure and `.describe()` for summary statistics.

Data Cleaning

Imputed missing values in `Review Rating` using median ratings by product category. Dropped redundant `Promo Code Used` column.

01

Column Standardization

Rename columns to snake case for readability

02

Feature Engineering

Create age groups and purchase frequency metrics

03

Database Integration

Load cleaned data into PostgreSQL

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SQL Analysis: Revenue Insights

Gender Revenue Split

Female: $75,191

Male: $157,890

Male customers generate 2.1x more revenue

Age Group Performance

Young Adult: $62,143

Middle-aged: $59,197

Adult: $55,978

Senior: $55,763

Shipping Comparison

Standard: $58.46 avg

Express: $60.48 avg

Express shipping correlates with higher spend

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Product Performance Analysis

Top-Rated Products

1. Gloves (3.86 avg)
2. Sandals (3.84 avg)
3. Boots (3.82 avg)
4. Hat (3.80 avg)
5. Skirt (3.78 avg)

Most Discounted Products

1. Hat (50.00%)
2. Sneakers (49.66%)
3. Coat (49.07%)
4. Sweater (48.17%)
5. Pants (47.37%)

High-discount products like hats and sneakers may indicate clearance strategy or competitive pricing. Top-rated items show strong customer satisfaction across accessories and footwear categories.

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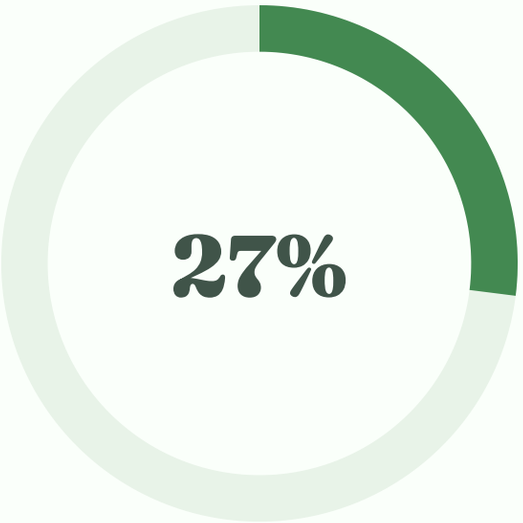
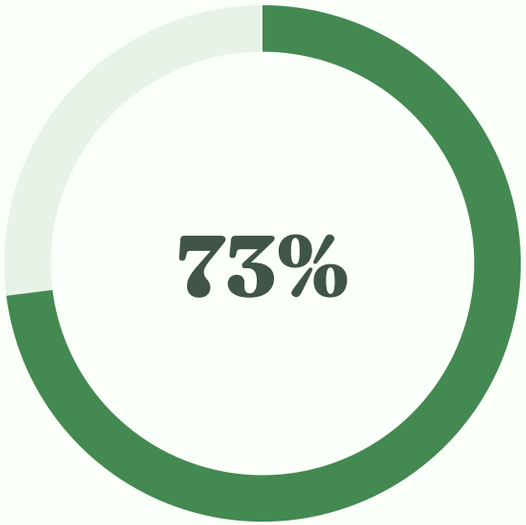
Subscription & Customer Segmentation



Subscribers

1,053 customers with active subscriptions



Non-Subscribers

2,847 customers without subscriptions

Loyal

3,116 customers with consistent purchase history

Returning

701 customers with multiple purchases

New

83 customers with single transactions

Subscribers show slightly lower average spend ($59.49 vs $59.87) but generate $62,645 total revenue. 958 repeat buyers (over 5 purchases) indicate strong retention potential.

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Top Products by Category

Accessories

- Jewelry (171 orders)
- Sunglasses (161 orders)
- Belt (161 orders)

Clothing

- Blouse (171 orders)
- Pants (171 orders)
- Shirt (169 orders)

Footwear

- Sandals (160 orders)
- Shoes (150 orders)
- Sneakers (145 orders)

Outerwear

- Jacket (163 orders)
- Coat (161 orders)

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Power BI Dashboard

Interactive KPIs

3.9K customers, $59.76 average purchase, 3.75 average rating

Visual Analytics

Revenue by category, sales by age group, subscription status breakdown

Segment Filters

Drill down by gender, category, shipping type, and subscription status

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Key Business Insights

Revenue Gap

Male customers generate 2.1x more revenue than female customers

Discount Strategy

50% of hat purchases discounted–balance sales with margin control

Loyalty Base

3,116 loyal customers represent strong retention foundation

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Strategic Recommendations

Boost Subscriptions

Promote exclusive benefits and incentives to increase subscriber base from current 27%

Enhance Loyalty Programs

Reward repeat buyers to accelerate movement into loyal customer segment

Optimize Discount Policy

Balance sales boosts with margin control–review 50% discount rate on hats

Strategic Product Positioning

Highlight top-rated products (gloves, sandals) and best-sellers in marketing campaigns

Targeted Marketing

Focus on high-revenue young adult segment and express-shipping customers

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